

ANISH NILESH RANE

AI/ML Engineer — Industrial Systems, Predictive Maintenance & Digital Twins

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years bridging applied machine learning and heavy industrial engineering at one of the world's largest diesel engine and hydraulics OEMs. Built and deployed production RAG search engines, LSTM predictive-maintenance models and LangChain-based agentic workflows concurrently with progressive hands-on roles in Hydraulics Sales, Technical-Commercial Engineering and Diesel Engine Application Engineering. Delivered measurable outcomes 60% accuracy gains in technical spec matching, 20–40% efficiency and downtime improvements and \$1.2M+ in incremental revenue impact.

SKILLS

- **Applied AI for Industrial Systems:** Agentic AI & multi-agent RL, RAG architectures, LLM orchestration (LangChain, LangGraph, Llama 3.1, Qwen2, Gemma2, GPT-4), predictive maintenance, digital twins, sensor-fusion anomaly detection, NLP, computer vision.
- **ML Engineering & Deployment:** Python, SQL, PyTorch, TensorFlow, Scikit-learn, LSTM/CNN/Conv3D/ConvLSTM, ChromaDB & vector search, model quantization & fine-tuning, AWS (SageMaker, Lambda), GCP, Azure ML, Docker, Kubernetes.
- **Industrial Domain Expertise:** Hydraulic systems (open/closed-loop pumps, motors, valves, controllers), diesel engine application engineering (duty-cycle, power/torque mapping, radiator & air-filter selection), CATIA V5, SolidWorks, Windchill/PLM, Teamcenter.
- **Commercial Fluency:** Technical sales cycle ownership, OEM relationship management (SPOC), technical-commercial evaluation, product growth & marketing.

PROFESSIONAL EXPERIENCE

Weichai Power — Pune, India

Diesel engines and hydraulics OEMs for construction and commercial machinery | May 2023 – Present

Application Engineer – Diesel Engines

Jan 2026 – Present

- Lead engine selection for OEM inquiries, mapping duty-cycle, power and torque requirements to the right engine platform including radiator and air-filter selection.
- Perform data logging and fault-finding on fielded engines, reviewing engineering drawings to diagnose and resolve performance issues.
- Coordinated field failure analysis for a fleet of 25 heavy-duty dump trucks using diagnostic tool DiagSmart pinpointing root causes of vehicular damage through historical data analytics.
- Deployed a standardized field validation checklist across construction and industrial applications (dump trucks, crawler cranes, wheel loaders, excavators, truck cranes, railway systems) ensuring 100% compliance.
- Developed standardized Long Block BOMs for core engine families (WP4.6, WP8, WP13) with Sales and Service teams aligning configurations with market demand.

Technical-Commercial Engineer

Sep 2024 – Jan 2026

- Owned the technical-commercial evaluation of hydraulic systems, running calculations and selecting the right pumps, motors and accessories to match customer application requirements.
- Authored and finalized Technical Agreements with OEMs defining scope of supply and application-specific requirements.
- Served as Single Point of Contact (SPOC) for 4 major OEMs streamlining technical coordination and rapid issue resolution.
- Conducted design reviews with OEMs to optimize engine compartment layouts (radiator, CAC, air intake, after-treatment) maximizing serviceability within tight spatial constraints.
- Led marketing initiatives for product growth exhibitions, promotional materials and LinkedIn company-page management alongside technical-commercial responsibilities.

Sales Engineer – Hydraulics

May 2023 – Aug 2024

- Managed the end-to-end sales cycle for closed-loop and open-loop hydraulic pumps, motors, valves and system accessories from initial customer consultation through deal closure.
- Built product and application knowledge of heavy-duty hydraulic systems for industrial and construction vehicles forming the technical foundation for later technical-commercial and engineering roles.

Built and shipped production AI/ML systems throughout all three roles above, applying GenAI and predictive modeling directly to Weichai's engineering and sales operations.

- Architected a production-grade RAG search engine (Qwen2 0.5B, ChromaDB, semantic retrieval) matching technical engine specifications to customer requirements boosting match accuracy by 60% and reducing sales-to-engineering handoff time.
- Built LSTM-based predictive maintenance models with real-time anomaly detection across 25 industrial sensors and hydraulic subsystems cutting unplanned downtime by ~20%.
- Designed agent-based AI workflows using LangChain and open-source LLMs (Llama 3.1, Qwen2, Gemma2) with tool orchestration, memory and planning for manufacturing use cases implemented tracing/logging for agent-decision transparency.
- Developed a production ML model predicting vehicle-level fuel efficiency and an enterprise job-site fuel-consumption algorithm reaching 97% accuracy.
- Built Gemma2-powered demand forecasting models on AWS using 2+ years of historical data optimizing \$0.8M+ in annual procurement costs.
- Engineered AI-driven digital twin simulations integrating CATIA/SolidWorks CAD models with open-source LLMs (CAD-Llama) for rapid design validation.
- Automated invoice and technical-document workflows with NLP accelerating delivery cycles by 40% and eliminating manual data-entry errors.
- Delivered LLM-generated sales insights for mobile crusher deployments contributing to \$1.2M+ incremental revenue by exceeding sales targets by 25%.

PROJECTS

LLM-Enhanced Multi-Agent Reinforcement Learning for Autonomous Driving — *MSc Dissertation, Liverpool John Moores University (2026)*

- Designed a framework where an LLM assigns semantic driving roles (yield, merge, block, proceed) offline during training only fusing them into a latent role embedding those conditions a decentralized multi-agent policy avoiding runtime LLM latency entirely.
- Architecture combined centralized-training/decentralized-execution multi-agent RL, inter-agent attention and an analytical safety shield benchmarked across 9 configurations (rule-based controller, standard RL baseline, proposed framework, 6 ablations) over 8 simulated scenarios with density, weather, incident and topological distribution shift.
- Improved fleet speed by 11.6% and cut collision rate by 65.6% versus a standard RL baseline reproducible across 6 independently seeded reruns identified the safety shield (not the LLM alone) as the dominant contributor to the safety gain.

RAG-Based Chatbot for Life Insurance (Python, GPT-3.5 Turbo, ChromaDB, LangChain) — 92% accuracy across 1,000+ documents with sub-2-second query response-built citation-accuracy evaluation framework.

Gesture Recognition with 3D-CNN (TensorFlow, Conv3D, ConvLSTM) — 78% validation accuracy on 15K video samples 30% training-time reduction via tuning and augmentation.

EDUCATION

Master's Degree, Machine Learning & Artificial Intelligence

Liverpool John Moores University, UK

Expected 2027

Dissertation: "LLM-Enhanced Multi-Agent Reinforcement Learning for Autonomous Driving"

Postgraduate Diploma, Machine Learning & AI — First-Class with Distinction

International Institute of Information Technology (IIIT), Bangalore

2023 – 2025

Bachelor of Engineering, Mechanical Engineering (Honours: Electric Vehicles) — First-Class with Distinction

Savitribai Phule Pune University

2020 – 2023

CERTIFICATIONS

- Google Startup School: Prompt to Prototype — Google for Startups
- S.O.L.I.D Principles & Docker/Kubernetes — Scaler
- Eicher Trucks – ALT-MDTAA & MHT-DTT3, Drivetrain Integration
- Garrett eBadge Series — Bronze to Platinum + Racing & Performance
- Power BI — Alison (2025) | Python & SQL — HackerRank